Math 163 — Exam 1B — Fully Worked Solutions

J-number 00499317 — every diagram drawn, every computation shown longhand

0. Setup — deriving your personal constants

0.1 Digits of the J-number

J-number = 00499317. Reading left to right, $J_1, \ldots, J_8$ are just the 8 digits:

k	1	2	3	4	5	6	7	8
J_k	0	0	4	9	9	3	1	7

0.2 $r_k := J_k \bmod 5$ — **long division (bus-stop form)**

D $r_k :=$ the remainder obtained upon dividing J_k by 5.

$$\begin{array}{r|l} & 0 \\ \hline 5 & 0 \\ & -0 \\ \hline & 0 \end{array} \quad J_1 \qquad \begin{array}{r|l} & 0 \\ \hline 5 & 0 \\ & -0 \\ \hline & 0 \end{array} \quad J_2 \qquad \begin{array}{r|l} & 0 \\ \hline 5 & 4 \\ & -0 \\ \hline & 4 \end{array} \quad J_3 \qquad \begin{array}{r|l} & 1 \\ \hline 5 & 9 \\ & -5 \\ \hline & 4 \end{array} \quad J_4$$

$$\begin{array}{r|l} & 1 \\ \hline 5 & 9 \\ & -5 \\ \hline & 4 \end{array} \quad J_5 \qquad \begin{array}{r|l} & 0 \\ \hline 5 & 3 \\ & -0 \\ \hline & 3 \end{array} \quad J_6 \qquad \begin{array}{r|l} & 0 \\ \hline 5 & 1 \\ & -0 \\ \hline & 1 \end{array} \quad J_7 \qquad \begin{array}{r|l} & 1 \\ \hline 5 & 7 \\ & -5 \\ \hline & 2 \end{array} \quad J_8$$

The remainder upon dividing $J_1 = 0$ by 5 is 0; hence $r_1 = 0$. The remainder upon dividing $J_2 = 0$ by 5 is 0; hence $r_2 = 0$. The remainder upon dividing $J_3 = 4$ by 5 is 4; hence $r_3 = 4$. The remainder upon dividing $J_4 = 9$ by 5 is 4; hence $r_4 = 4$. The remainder upon dividing $J_5 = 9$ by 5 is 4; hence $r_5 = 4$. The remainder upon dividing $J_6 = 3$ by 5 is 3; hence $r_6 = 3$. The remainder upon dividing $J_7 = 1$ by 5 is 1; hence $r_7 = 1$. The remainder upon dividing $J_8 = 7$ by 5 is 2; hence $r_8 = 2$.

$$r_1 = 0, \quad r_2 = 0, \quad r_3 = 4, \quad r_4 = 4, \quad r_5 = 4, \quad r_6 = 3, \quad r_7 = 1, \quad r_8 = 2$$

0.3 $Prm(k)$ **for** $k = 1, \ldots, 8$ — **trial division**

To test if k is prime, divide by every integer from 2 up to $\lfloor \sqrt{k} \rfloor$; if none divide evenly, k is prime ($k \geq 2$). $k = 1$ needs no trial division at all — by definition 1 is *never* prime.

$k = 1:$ not prime (by definition, 1 is excluded)
$k = 2:$ $\lfloor \sqrt{2} \rfloor = 1$, no divisor to test ⇒ **prime**
$k = 3:$ $\lfloor \sqrt{3} \rfloor = 1$, no divisor to test ⇒ **prime**
$k = 4:$ $\lfloor \sqrt{4} \rfloor = 2$, $4 \div 2 = 2$ r0 ⇒ divisible ⇒ not prime
$k = 5:$ $\lfloor \sqrt{5} \rfloor = 2$, $5 \div 2 = 2$ r1 ⇒ no divisor found ⇒ **prime**
$k = 6:$ $\lfloor \sqrt{6} \rfloor = 2$, $6 \div 2 = 3$ r0 ⇒ divisible ⇒ not prime
$k = 7:$ $\lfloor \sqrt{7} \rfloor = 2$, $7 \div 2 = 3$ r1 ⇒ no divisor found ⇒ **prime**
$k = 8:$ $\lfloor \sqrt{8} \rfloor = 2$, $8 \div 2 = 4$ r0 ⇒ divisible ⇒ not prime

Recording each test as a circled-choice grid, matching the exam's own $Prm(k)$ table format:

k	1	2	3	4	5	6	7	8
J_k	0	0	4	9	9	3	1	7
$Prm(k) = Y$	N	[Y]	[Y]	N	[Y]	N	[Y]	N
$Prm(k) = N$	[N]	N	N	[N]	N	[N]	N	[N]

So $\{k \in 1..8 \mid Prm(k)\} = \{2, 3, 5, 7\}$.

0.4 Building E, s, b, A, B, R

R For each of the sets defined below, write the elements in increasing order.

D Define $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

D Define $J := \{J_k \in 0..9 \mid k \in 1..8\} = \{0, 1, 3, 4, 7, 9\}$ (the eight digits $0, 0, 4, 9, 9, 3, 1, 7$, deduplicated and sorted increasing).

D Define $\forall n \in \mathbb{N} \quad Prm(n) :\Leftrightarrow n$ is a prime number.

D Define

$$E \;:=\; \{J_k \in 0..9 \mid Prm(k)\} \;=\; \{J_2, J_3, J_5, J_7\} \;=\; \{0, 4, 9, 1\} \;=\; \{0, 1, 4, 9\}$$

using the $Prm(k)$ determination from **0.3**: the prime indices are $k = 2, 3, 5, 7$, giving $J_2 = 0,\ J_3 = 4,\ J_5 = 9,\ J_7 = 1$; sorted increasing, $E = \{0, 1, 4, 9\}$.

D $s \;:=\; \mathrm{mnm}_{k \in 1..8}\{J_k \in 0..9 \mid J_k \neq 0\}$.

$$\{J_3, J_4, J_5, J_6, J_7, J_8\} = \{4, 9, 9, 3, 1, 7\}, \qquad \text{minimum attained at } k = 7 \ (J_7 = 1) \;\Rightarrow\; s = 1$$

D $b \;:=\; \mathrm{mxm}_{k \in 1..8}\{J_k \in 0..9 \mid k \in 1..8\} + 1$. All eight digits are $\{0, 0, 4, 9, 9, 3, 1, 7\}$; their maximum is 9 (attained at $k = 4$ and $k = 5$), so $b = 9 + 1 = 10$.

D Define

$$A \;:=\; \{0, s, b\} \;=\; \{0, 1, 10\}$$

D Define

$$B \;:=\; \{J_k \in 0..9 \mid Prm(k)\} \;=\; \{0, 1, 4, 9\} \quad (= E, \text{ same defining formula})$$

R Note that in a set an element may occur exactly once, so there are no repeated elements in either A or B above. A and B recur throughout the exam, so any error in these two sets propagates to every later answer.

D Define a relation $R \subseteq \{0, 1, 2, 3, 4\} \times \{0, 1, 2, 3, 4\} =: \mathcal{U}$ by

$$R \;:=\; \{(0, 0)\} \cup \bigcup_{k \in 1..4} \{(r_{1+k}, r_{2+k})\} \;=\; \{(0, 0)\} \cup \{(r_2, r_3), (r_3, r_4), (r_4, r_5), (r_5, r_6)\}$$

Substituting $r_2 = 0,\ r_3 = 4,\ r_4 = 4,\ r_5 = 4,\ r_6 = 3$ from **0.2**:

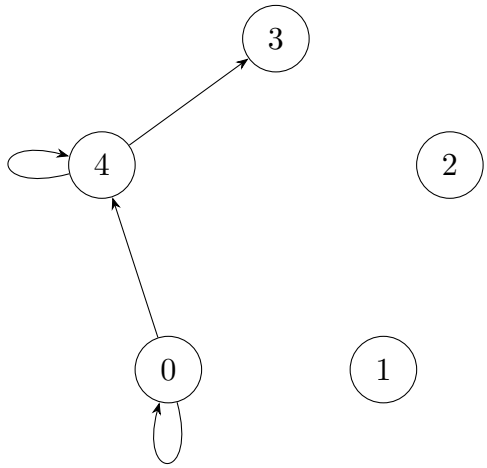
$$R \;=\; \{(0, 0)\} \cup \{(0, 4), (4, 4), (4, 4), (4, 3)\} \;=\; \{(0, 0), (0, 4), (4, 3), (4, 4)\}$$

(the repeated pair $(4, 4)$ only counts once, since R is a *set*.)

1(abc) — the relation R on $\{0,1,2,3,4\}$

(i) Digraph $DG(R)$

1abc.i

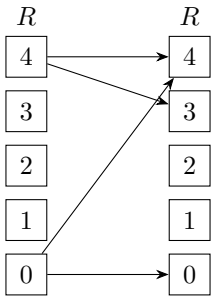


Vertices $0,\ldots,4$ drawn as a pentagon. Self-loops at 0 and 4 (from $(0,0),(4,4)$); arrow $0\to 4$; arrow $4\to 3$; vertices $1,2$ have no edges at all (isolated).

(ii)–(iii) Matrices

$R^{op}=\{(0,0),(3,4),(4,0),(4,4)\}$. Per the exam's own rule ("in arrow-diagrams, put the numbers in rectangles rather than ovals"), R and R^{op} are first drawn as two-column **arrow-diagrams** $AD(R)$, $AD(R^{op})$ (domain copy of $\{0,..,4\}$ on the left, codomain copy on the right, elements listed top-to-bottom $4,3,2,1,0$) — this is the scratch work used to read off each matrix entry.

1abc.ii.0



$AD(R)$

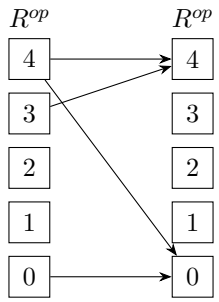
1abc.ii.1

$M(R)$, read off $AD(R)$ (row = source, column = target; 1 if an arrow goes source→target, else 0): arrows are $L0\to R0$, $L0\to R4$, $L4\to R3$, $L4\to R4$, so row 0 has 1s at columns $0,4$ and row 4 has 1s at columns $3,4$; all other rows are 0. As a headed grid, then as a bracket matrix:

$M(R)$	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	0	0	0	1	1

$$M(R)=\begin{bmatrix}1&0&0&0&1\\0&0&0&0&0\\0&0&0&0&0\\0&0&0&0&0\\0&0&0&1&1\end{bmatrix}$$

1abc.iii.0



$AD(R^{op})$

1abc.iii.1

$M(R^{op})$, read off $AD(R^{op})$: arrows $L0{\to}R0$, $L3{\to}R4$, $L4{\to}R0$, $L4{\to}R4$, so row 0 has a 1 at column 0, row 3 has a 1 at column 4, row 4 has 1s at columns $0, 4$:

$M(R^{op})$	0	1	2	3	4
0	1	0	0	0	0
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	1
4	1	0	0	0	1

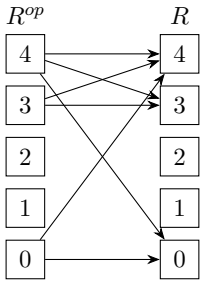
$$M(R^{op}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Rows/columns indexed $0, 1, 2, 3, 4$ in order throughout.

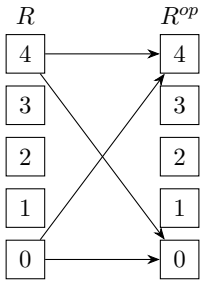
(iv) Matrix identities — shown entry by entry

1abc.iv.0

The two composed relations as their own arrow-diagrams (numbers in rectangles, per the exam's rule):



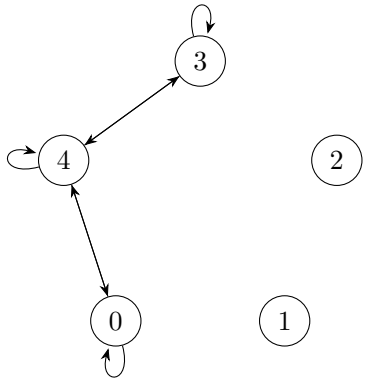
$AD(R \circ R^{op})$



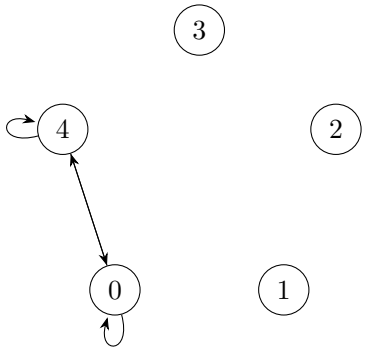
$AD(R^{op} \circ R)$

Composition convention: $R \circ S := \{(a, c) \mid \exists b,\ (a, b) \in S \wedge (b, c) \in R\}$, so $M(R \circ S) = M(S) \cdot M(R)$ (apply S first).

$$R \circ R^{op} = \{(0,0), (0,4), (3,3), (3,4), (4,0), (4,3), (4,4)\}, \qquad R^{op} \circ R = \{(0,0), (0,4), (4,0), (4,4)\}$$



$DG(R \circ R^{op})$



$DG(R^{op} \circ R)$

1abc.iv.1

Identity 1: $\big(M(R \circ R^{op})\big)(0,1) = \big((M(R^{op}))(M(R))\big)(0,1)$.

$$\big((M(R^{op}))(M(R))\big) = \left(\begin{bmatrix} 1&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&0&1 \\ 1&0&0&0&1 \end{bmatrix} \begin{bmatrix} 1&0&0&0&1 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&1 \end{bmatrix} \right) = \begin{bmatrix} a&b&c&d&e \\ f&g&h&i&j \\ k&l&m&n&o \\ p&q&r&s&t \\ u&v&w&x&y \end{bmatrix}$$

Every one of the 25 entries, computed as a full dot-product sum (row of $M(R^{op})$ against column of $M(R)$), labeled $a, \ldots, y$ left-to-right, top-to-bottom:

$a := (1)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 1$
$b := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$c := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$d := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$e := (1)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 1$
$f := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$g := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$h := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$i := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$j := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$k := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$l := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$m := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$n := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$o := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$p := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$q := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$r := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$s := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(1) = 1$
$t := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(1)(1) = 1$
$u := (1)(1)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 1$
$v := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$w := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$x := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(1) = 1$
$y := (1)(1)+(0)(0)+(0)(0)+(0)(0)+(1)(1) = 2$

$$\big((M(R^{op}))(M(R))\big)(0,1) = \begin{bmatrix} 1&0&0&0&1 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&1 \\ 1&0&0&1&\mathbf{2} \end{bmatrix} (0,1) = \begin{bmatrix} 1&0&0&0&1 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&1 \\ 1&0&0&1&1 \end{bmatrix}$$

1abc.iv.2

As a headed grid, then bracket, $M(R \circ R^{op})$ (read directly off $DG(R \circ R^{op})$ / $AD(R \circ R^{op})$):

$M(R \circ R^{op})$	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	1	1
4	1	0	0	1	1

$$M(R \circ R^{op}) = \begin{bmatrix} 1&0&0&0&1 \\ 0&0&0&0&0 \\ 0&0&0&0&0 \\ 0&0&0&1&1 \\ 1&0&0&1&1 \end{bmatrix}$$

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

1abc.iv.3

Identity 2: $\big(M(R^{op} \circ R)\big)(0,1) = \big((M(R))(M(R^{op}))\big)(0,1)$.

$$\big((M(R))(M(R^{op}))\big) = \left(\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \right) = \begin{bmatrix} a & b & c & d & e \\ f & g & h & i & j \\ k & l & m & n & o \\ p & q & r & s & t \\ u & v & w & x & y \end{bmatrix}$$

$a := (1)(1)+(0)(0)+(0)(0)+(0)(0)+(1)(1) = 2$ $b := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$c := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$ $d := (1)(0)+(0)(0)+(0)(0)+(0)(0)+(1)(0) = 0$
$e := (1)(0)+(0)(0)+(0)(0)+(0)(1)+(1)(1) = 1$ $f := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$g := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$ $h := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$i := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$ $j := (0)(0)+(0)(0)+(0)(0)+(0)(1)+(0)(1) = 0$
$k := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$ $l := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$m := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$ $n := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$o := (0)(0)+(0)(0)+(0)(0)+(0)(1)+(0)(1) = 0$ $p := (0)(1)+(0)(0)+(0)(0)+(0)(0)+(0)(1) = 0$
$q := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$ $r := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$
$s := (0)(0)+(0)(0)+(0)(0)+(0)(0)+(0)(0) = 0$ $t := (0)(0)+(0)(0)+(0)(0)+(0)(1)+(0)(1) = 0$
$u := (0)(1)+(0)(0)+(0)(0)+(1)(0)+(1)(1) = 1$ $v := (0)(0)+(0)(0)+(0)(0)+(1)(0)+(1)(0) = 0$
$w := (0)(0)+(0)(0)+(0)(0)+(1)(0)+(1)(0) = 0$ $x := (0)(0)+(0)(0)+(0)(0)+(1)(0)+(1)(0) = 0$
$y := (0)(0)+(0)(0)+(0)(0)+(1)(1)+(1)(1) = 2$

$$\big((M(R))(M(R^{op}))\big)(0,1) = \begin{bmatrix} \mathbf{2} & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & \mathbf{2} \end{bmatrix} (0,1) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1abc.iv.4

As a headed grid, then bracket, $M(R^{op} \circ R)$:

$M(R^{op} \circ R)$	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	1	0	0	0	1

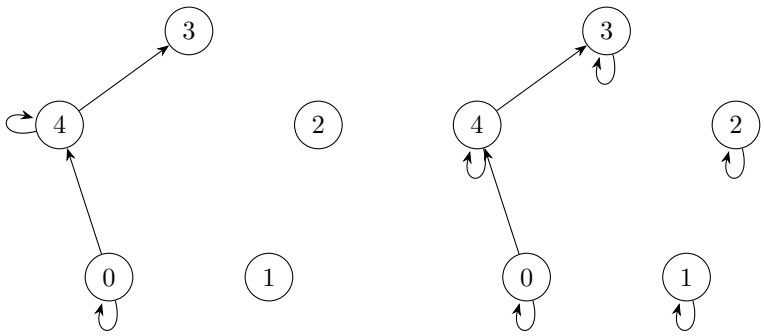
$$M(R^{op} \circ R) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

(v) *FlxClsr*(R) — **reflexive closure**

1abc.v.0



$DG(R)$ $DG(FlxClsr(R))$

1abc.v.1

FlxClsr(R), read directly off $DG(FlxClsr(R))$ by adding a self-loop at every vertex that lacks one:

$$FlxClsr(R) = \{(0,0),(0,4),(1,1),(2,2),(3,3),(4,3),(4,4)\}$$

1abc.v.2

$\big(M(FlxClsr(R))\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	0	1
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	1	0
4	0	0	0	1	1

$$\big(M(FlxClsr(R))\big)(0,1) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

1abc.v.3

$\Delta_{\{0,..,4\}} = \{(0,0),(1,1),(2,2),(3,3),(4,4)\}$

1abc.v.4

$$\begin{aligned} R \cup \big(\Delta_{\{0,..,4\}}\big) &= \{(0,0),(0,4),(4,3),(4,4)\} \cup \{(0,0),(1,1),(2,2),(3,3),(4,4)\} \\ &= \{(0,0),(0,4),(1,1),(2,2),(3,3),(4,3),(4,4)\} \end{aligned}$$

1abc.v.5

$\big(M(R \cup (\Delta_{\{0,..,4\}}))\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	0	1
1	0	1	0	0	0
2	0	0	1	0	0
3	0	0	0	1	0
4	0	0	0	1	1

$$\big(M(R \cup (\Delta_{\{0,..,4\}}))\big)(0,1) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

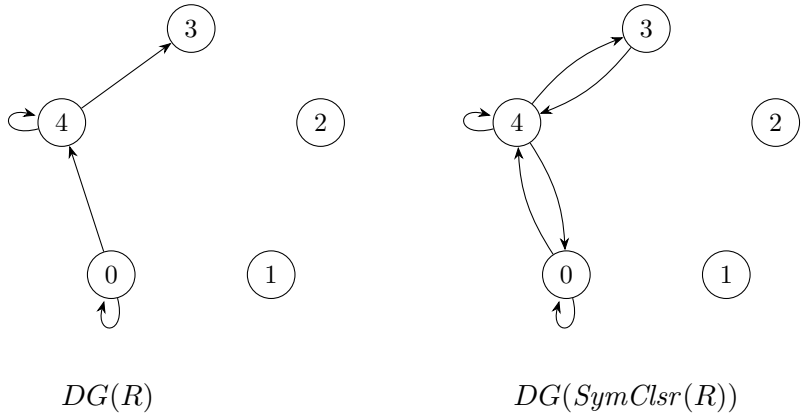
Prove: $\big(M(FlxClsr(R))\big)(0,1) = \big(M(R \cup (\Delta_{\{0,..,4\}}))\big)(0,1)$.

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

(vi) $SymClsr(R)$ — **symmetric closure**

1abc.vi.0



$DG(R)$ $DG(SymClsr(R))$

1abc.vi.1

$SymClsr(R)$, read off $DG(SymClsr(R))$ by adding the reverse of every edge that lacks a matching reverse:

$$SymClsr(R) = \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\}$$

1abc.vi.2

$\big(M(SymClsr(R))\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	1
4	1	0	0	1	1

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

1abc.vi.3

$$\begin{aligned} R \cup (R^{op}) &= \{(0,0),(0,4),(4,3),(4,4)\} \cup \{(0,0),(3,4),(4,0),(4,4)\} \\ &= \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\} \end{aligned}$$

1abc.vi.4

$\big(M(R \cup (R^{op}))\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	1
4	1	0	0	1	1

$$\big(M(R \cup (R^{op}))\big)(0,1) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

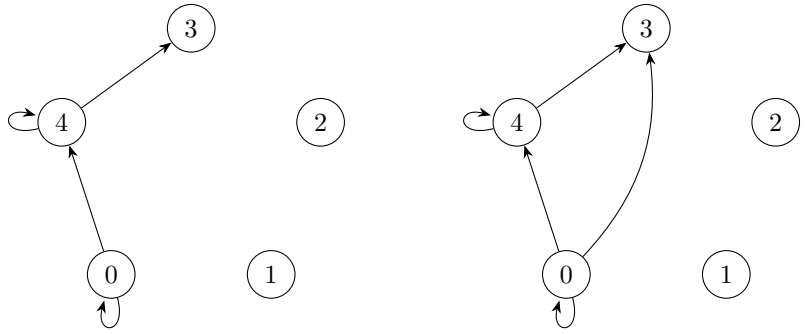
Prove: $\big(M(SymClsr(R))\big)(0,1) = \big(M(R \cup (R^{op}))\big)(0,1)$.

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

(vii) $TrnsClsr(R)$ — **transitive closure, powers via arrow-diagram chase**

1abc.vii.0

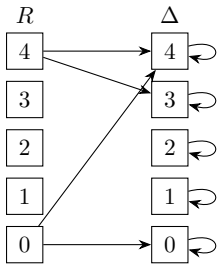


$DG(R)$ $DG(TrnsClsr(R))$

We compute $R^{\circ k} := R^{\circ(k-1)} \circ R$ for $k = 1, \ldots, 5$ (with $R^{\circ 0} := \Delta_{\{0,..,4\}}$), each power found by an **arrow-diagram chase** rather than numeric matrix multiplication: draw R (left column → middle column) and $R^{\circ(k-1)}$ (middle column → right column) side by side, then read off $R^{\circ k}$ as every left-to-right pair reachable via some middle vertex.

1abc.vii.1

$k = 1$: $R^{\circ 1} = R^{\circ 0} \circ R = \Delta_{\{0,..,4\}} \circ R$.

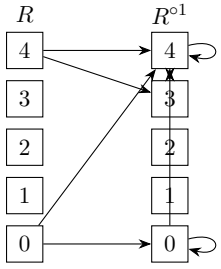


$AD(\Delta_{\{0,..,4\}} \circ R)$ — chasing through the identity changes nothing

$R^{\circ 1} = \{(0,0), (0,4), (4,3), (4,4)\}$

1abc.vii.2

$k = 2$: $R^{\circ 2} = R^{\circ 1} \circ R$.



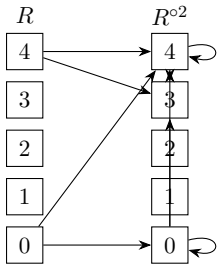
$AD(R^{\circ 1} \circ R)$

Chasing: $0 \to 0 \to \{0,4\}$ and $0 \to 4 \to \{3,4\}$, so 0 reaches $\{0,3,4\}$; $4 \to 3 \to \{\}$ (no arrows leave 3 in $R^{\circ 1}$) and $4 \to 4 \to \{3,4\}$, so 4 reaches $\{3,4\}$.

$$R^{\circ 2} = \{(0,0), (0,3), (0,4), (4,3), (4,4)\}$$

1abc.vii.3

$k = 3$: $R^{\circ 3} = R^{\circ 2} \circ R$.



$AD(R^{\circ 2} \circ R)$

Chasing: $0 \to 0 \to \{0,3,4\}$ and $0 \to 4 \to \{3,4\}$, so 0 reaches $\{0,3,4\}$; $4 \to 3 \to \{\}$ and $4 \to 4 \to \{3,4\}$, so 4 reaches $\{3,4\}$ — identical to $R^{\circ 2}$.

$$R^{\circ 3} = \{(0,0),(0,3),(0,4),(4,3),(4,4)\} = R^{\circ 2}$$

No new pairs appeared, so the chase has **stabilized**. Composing an already-stable relation with R again reproduces the same result, so $R^{\circ 4} = R^{\circ 5} = R^{\circ 2}$ as well, without needing to redo the chase.

1abc.vii.4

$$R^{\circ 4} = R^{\circ 3} \circ R = R^{\circ 2} = \{(0,0),(0,3),(0,4),(4,3),(4,4)\}$$
$$R^{\circ 5} = R^{\circ 4} \circ R = R^{\circ 2} = \{(0,0),(0,3),(0,4),(4,3),(4,4)\}$$

1abc.vii.5

$$\bigcup_{k=1}^{5} R^{\circ k} = R^{\circ 1} \cup R^{\circ 2} \cup R^{\circ 3} \cup R^{\circ 4} \cup R^{\circ 5} = \{(0,0),(0,3),(0,4),(4,3),(4,4)\}$$

Why stopping at $k = 5$ is enough (pigeonhole): $\nu(\{0,..,4\}) = 5$. Any directed path with more than 5 edges must revisit a vertex, giving a shorter path (delete the cycle) with the same endpoints. So every reachable pair is already reachable by a path of length ≤ 5.

1abc.vii.6

$\big(M(\mathit{TrnsClsr}(R))\big)(0,1)$ and $\big(M(\bigcup_{k=1}^{5} R^{\circ k})\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	1	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	0
4	1	0	0	1	1

$$= \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

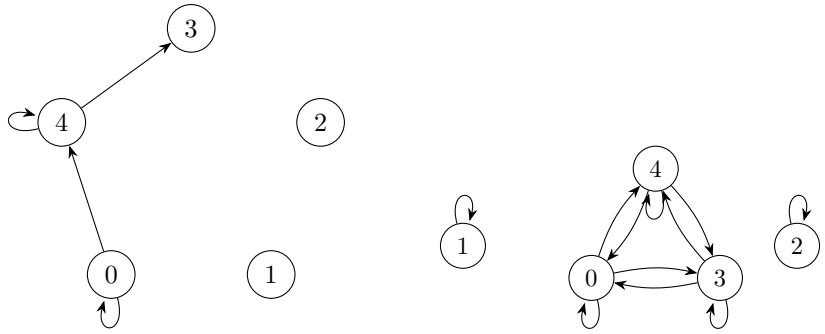
Prove: $\big(M(\mathit{TrnsClsr}(R))\big)(0,1) = \big(M(\bigcup_{k=1}^{5} R^{\circ k})\big)(0,1)$.

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

(viii) $EqvClsr(R)$ — **equivalence closure, equivalence classes, quotient**

1abc.viii.0



$DG(R)$ $DG(EqvClsr(R))$

1abc.viii.1

$EqvClsr(R)$, read directly off $DG(EqvClsr(R))$ (the smallest reflexive, symmetric, transitive relation containing R — every vertex gets a loop, every edge gets its reverse, and $\{0,3,4\}$ close up into a clique since $0,3,4$ are all mutually reachable through R):

$$EqvClsr(R) = \{(0,0),(0,3),(0,4),(1,1),(2,2),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\}$$

1abc.viii.2

$\big(M(EqvClsr(R))\big)(0,1)$:

	0	1	2	3	4
0	1	0	0	1	1
1	0	1	0	0	0
2	0	0	1	0	0
3	1	0	0	1	1
4	1	0	0	1	1

$$\big(M(EqvClsr(R))\big)(0,1) = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

1abc.viii.3

Equivalence class of each element under $EqvClsr(R)$ (row x of the matrix above, read as a set):

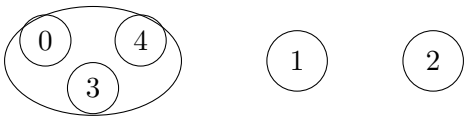
$$EqvCls(0, EqvClsr(R)) = \{0,3,4\}, \quad EqvCls(1, EqvClsr(R)) = \{1\}, \quad EqvCls(2, EqvClsr(R)) = \{2\},$$

$$EqvCls(3, EqvClsr(R)) = \{0,3,4\}, \quad EqvCls(4, EqvClsr(R)) = \{0,3,4\}$$

1abc.viii.4

Quotient:

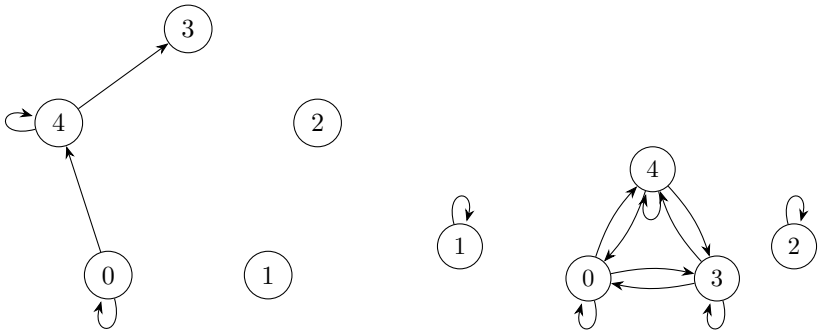
$$\frac{\{0,1,2,3,4\}}{EqvClsr(R)} = \big\{EqvCls(x, EqvClsr(R)) \in \mathcal{P}(0..4) \mid x \in \{0,1,2,3,4\}\big\} = \big\{\{0,3,4\},\{1\},\{2\}\big\}$$



The digraph $DG(EqvClsr(R))$ splits into 3 disconnected cliques $\{0,3,4\}$, $\{1\}$, $\{2\}$, one per equivalence class.

(ix) $EqvClsr(R)$ — **powers of** $Q := R \cup R^{op}$ **via arrow-diagram chase**

1abc.ix.0



$DG(R)$ $DG(EqvClsr(R))$

1abc.ix.1

$$\big(M(EqvClsr(R))\big)(0,1) = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} \text{ (from part (viii)).}$$

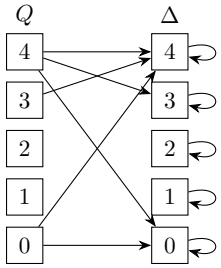
1abc.ix.2

$$R = \{(0,0),(0,4),(4,3),(4,4)\}, \qquad R^{op} = \{(0,0),(3,4),(4,0),(4,4)\}$$
$$Q := R \cup (R^{op}) = \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\}$$

We compute $Q^{\circ k} := Q^{\circ(k-1)} \circ Q$ for $k = 1, \ldots, 5$ (with $Q^{\circ 0} := \Delta_{\{0,..,4\}}$), again by **arrow-diagram chase**, the same method as part (vii).

1abc.ix.3

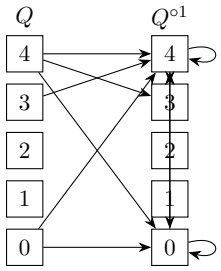
$k = 1$: $Q^{\circ 1} = Q^{\circ 0} \circ Q = \Delta_{\{0,..,4\}} \circ Q$.



$AD(\Delta_{\{0,..,4\}} \circ Q)$ — chasing through the identity changes nothing
$Q^{\circ 1} = \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\}$

1abc.ix.4

$k = 2$: $Q^{\circ 2} = Q^{\circ 1} \circ Q$.



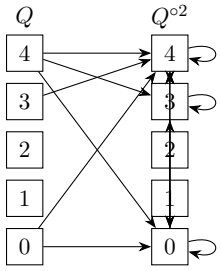
$AD(Q^{\circ 1} \circ Q)$

Chasing: $0 \to 0 \to \{0,4\}$, $0 \to 4 \to \{0,3,4\} \Rightarrow 0$ reaches $\{0,3,4\}$. $3 \to 4 \to \{0,3,4\} \Rightarrow 3$ reaches $\{0,3,4\}$. $4 \to 0 \to \{0,4\}$, $4 \to 3 \to \{4\}$, $4 \to 4 \to \{0,3,4\} \Rightarrow 4$ reaches $\{0,3,4\}$.

$$Q^{\circ 2} = \{(0,0),(0,3),(0,4),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\}$$

1abc.ix.5

$k = 3$: $Q^{\circ 3} = Q^{\circ 2} \circ Q$.



$AD(Q^{\circ 2} \circ Q)$

Chasing from 0, 3, 4 each lands on $\{0,3,4\}$ again — identical to $Q^{\circ 2}$.

$$Q^{\circ 3} = \{(0,0),(0,3),(0,4),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\} = Q^{\circ 2}$$

Stabilized: $Q^{\circ 4} = Q^{\circ 5} = Q^{\circ 2}$, by the same repeat-once-stable argument as part (vii).

1abc.ix.6

$$Q^{\circ 4} = Q^{\circ 2} = \{(0,0),(0,3),(0,4),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\}, \qquad Q^{\circ 5} = Q^{\circ 2}$$

1abc.ix.7

Why stopping at $k = 5$ is enough (pigeonhole), including $k = 0$ for reflexivity: $\nu(\{0,..,4\}) = 5$; any path needing more than 5 edges has a shorter equivalent (delete the revisited-vertex cycle), so checking $Q^{\circ 0}$ through $Q^{\circ 5}$ catches every pair. Note Q never relates vertex 1 or 2 to anything, so $(1,1)$ and $(2,2)$ enter only through $Q^{\circ 0} := \Delta$.

$$\bigcup_{k=0}^{5} \left((R \cup (R^{op}))^{\circ k}\right) = Q^{\circ 0} \cup Q^{\circ 1} \cup Q^{\circ 2} \cup Q^{\circ 3} \cup Q^{\circ 4} \cup Q^{\circ 5}$$

$$= \{(0,0),(1,1),(2,2),(3,3),(4,4)\} \cup \{(0,0),(0,4),(3,4),(4,0),(4,3),(4,4)\}$$

$$\cup \{(0,0),(0,3),(0,4),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\}$$

$$= \{(0,0),(0,3),(0,4),(1,1),(2,2),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)\}$$

1abc.ix.8

$\left(M(\bigcup_{k=0}^{5}(R \cup (R^{op}))^{\circ k})\right)(0,1)$:

	0	1	2	3	4
0	1	0	0	1	1
1	0	1	0	0	0
2	0	0	1	0	0
3	1	0	0	1	1
4	1	0	0	1	1

$$= \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

Prove: $\big(M(EqvClsr(R))\big)(0,1) = \big(M(\bigcup_{k=0}^{5}(R \cup (R^{op}))^{\circ k})\big)(0,1)$.

$$\text{LS} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \end{bmatrix} = \text{RS}$$

$$\text{LS} = \text{RS}$$

2a — injective/surjective functions $A \to B$

$A = \{0, 1, 10\}$, $\nu(A) = 3$. $B = \{0, 1, 4, 9\}$, $\nu(B) = 4$. Four *separate* propositions, one function-name each $(\alpha, \beta, \gamma, \delta)$, with the injective/surjective conjunct struck through wherever it's *negated*:

$$\text{(i) } \exists\alpha\big((Inj(\alpha)) \wedge (Srj(\alpha))\big), \quad \text{(ii) } \exists\beta\big((\cancel{Inj}(\beta)) \wedge (Srj(\beta))\big),$$

$$\text{(iii) } \exists\gamma\big((Inj(\gamma)) \wedge (\cancel{Srj}(\gamma))\big), \quad \text{(iv) } \exists\delta\big((\cancel{Inj}(\delta)) \wedge (\cancel{Srj}(\delta))\big)$$

2a.0

Lemmas used below (standard facts about finite functions):

$$\frac{InjFncs(A,B) \neq \{\}}{\nu(A) \leq \nu(B)} \qquad \frac{SrjFncs(A,B) \neq \{\}}{\nu(A) \geq \nu(B)}$$

Here $\nu(A) = 3 < 4 = \nu(B)$, so the right-hand lemma immediately gives $SrjFncs(A, B) = \{\}$: **no** function $A \to B$ is ever surjective, full stop. That one fact drives (i)–(iv) below.

(i) N — impossible

2a.i

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\alpha \in Fnc(A,B) \quad Inj(\alpha) \wedge Srj(\alpha)}{Srj(\alpha)} \quad I := Im(\alpha) \quad \nu(A) = 3,\ \nu(B) = 4}{\nu(I) = \nu(B) = 4}}{\nu(I) \leq \nu(A) = 3}}{4 \leq 3}}{\bot}}{\neg\exists\alpha \in Fnc(A,B)\big((Inj(\alpha)) \wedge (Srj(\alpha))\big)}$$

(ii) N — impossible

2a.ii

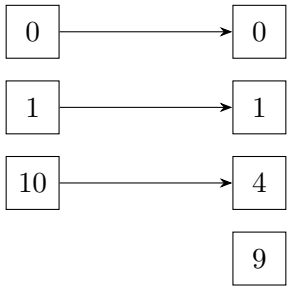
$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\beta \in Fnc(A,B) \quad Srj(\beta)}{Srj(\beta)} \quad I := Im(\beta) \quad \nu(A) = 3,\ \nu(B) = 4}{\nu(I) = \nu(B) = 4}}{\nu(I) \leq \nu(A) = 3}}{4 \leq 3}}{\bot}}{\neg\exists\beta \in Fnc(A,B)\big((\cancel{Inj}(\beta)) \wedge (Srj(\beta))\big)}$$

(iii) Y — witnessed

2a.iii

Since $\nu(A) = 3 \leq 4 = \nu(B)$, the lemma gives $InjFncs(A, B) \neq \{\}$, so injective functions $A \to B$ exist; and *every* function $A \to B$ automatically satisfies $\neg Srj$ (shown above), so any injective witness works. Take

$$\gamma : A \to B, \qquad \gamma(0) = 0, \quad \gamma(1) = 1, \quad \gamma(10) = 4$$



$Im(\gamma) = \{0, 1, 4\}$, all three outputs distinct $\Rightarrow$ $Inj(\gamma)$. $9 \notin Im(\gamma) \Rightarrow \neg Srj(\gamma)$. So $Inj(\gamma) \wedge \neg Srj(\gamma)$ holds:

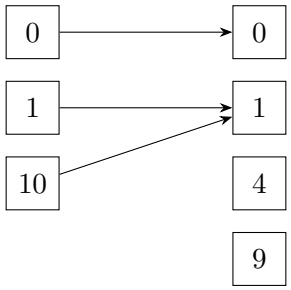
$$\therefore \exists \gamma \in Fnc(A, B)\big((Inj(\gamma)) \wedge (\cancel{Srj}(\gamma))\big) \qquad \checkmark$$

(iv) Y — witnessed

2a.iv

Every function $A \to B$ is automatically $\neg Srj$ (shown above), so we only need *some* non-injective δ. Take

$$\delta : A \to B, \qquad \delta(0) = 0, \quad \delta(1) = 1, \quad \delta(10) = 1$$



$\delta(1) = \delta(10) = 1$ with $1 \neq 10 \Rightarrow \neg Inj(\delta)$. $Im(\delta) = \{0, 1\} \neq B \Rightarrow \neg Srj(\delta)$. So $\neg Inj(\delta) \wedge \neg Srj(\delta)$ holds:

$$\therefore \exists \delta \in Fnc(A, B)\big((\cancel{Inj}(\delta)) \wedge (\cancel{Srj}(\delta))\big) \qquad \checkmark$$

(v)–(vi): image and maximal bijective restrictions of δ

2a.v-vi

$$Im(\delta) = \{0, 1\}$$

Since $\delta(1) = \delta(10) = 1$, any subset of A containing *both* 1 and 10 fails injectivity. The maximal subsets of A on which δ is injective are $\{0, 1\}$ and $\{0, 10\}$:



$$\delta|_{\{0,1\}} : \{0, 1\} \to \{0, 1\} \qquad \delta|_{\{0,10\}} : \{0, 10\} \to \{0, 1\}$$

Both restrictions are bijections onto $\{0, 1\}$. Count of maximal injective (adjusted-bijective) restrictions $= \mathbf{2}$.

2b — R (same parity) and S (order-or-divides)

$R := \{(j,k) \in J \times J \mid 2 \mid |j-k|\}$, $J = \{0,1,3,4,7,9\}$.

(i) **Y** $Eqv(R)$

2b.i

Exhaustive check of all 36 pairs of $J \times J$ (condition on top, membership conclusion below):

$$\begin{array}{cccccc}
\frac{Evn(0+0)}{(0,0)\in R} & \frac{Evn(|0-1|)}{(0,1)\notin R} & \frac{Evn(|0-3|)}{(0,3)\notin R} & \frac{Evn(|0-4|)}{(0,4)\in R} & \frac{Evn(|0-7|)}{(0,7)\notin R} & \frac{Evn(|0-9|)}{(0,9)\notin R} \\
\frac{Evn(|1-0|)}{(1,0)\notin R} & \frac{Evn(1+1)}{(1,1)\in R} & \frac{Evn(|1-3|)}{(1,3)\in R} & \frac{Evn(|1-4|)}{(1,4)\notin R} & \frac{Evn(|1-7|)}{(1,7)\in R} & \frac{Evn(|1-9|)}{(1,9)\in R} \\
\frac{Evn(|3-0|)}{(3,0)\notin R} & \frac{Evn(|3-1|)}{(3,1)\in R} & \frac{Evn(3+3)}{(3,3)\in R} & \frac{Evn(|3-4|)}{(3,4)\notin R} & \frac{Evn(|3-7|)}{(3,7)\in R} & \frac{Evn(|3-9|)}{(3,9)\in R} \\
\frac{Evn(|4-0|)}{(4,0)\in R} & \frac{Evn(|4-1|)}{(4,1)\notin R} & \frac{Evn(|4-3|)}{(4,3)\notin R} & \frac{Evn(4+4)}{(4,4)\in R} & \frac{Evn(|4-7|)}{(4,7)\notin R} & \frac{Evn(|4-9|)}{(4,9)\notin R} \\
\frac{Evn(|7-0|)}{(7,0)\notin R} & \frac{Evn(|7-1|)}{(7,1)\in R} & \frac{Evn(|7-3|)}{(7,3)\in R} & \frac{Evn(|7-4|)}{(7,4)\notin R} & \frac{Evn(7+7)}{(7,7)\in R} & \frac{Evn(|7-9|)}{(7,9)\in R} \\
\frac{Evn(|9-0|)}{(9,0)\notin R} & \frac{Evn(|9-1|)}{(9,1)\in R} & \frac{Evn(|9-3|)}{(9,3)\in R} & \frac{Evn(|9-4|)}{(9,4)\notin R} & \frac{Evn(|9-7|)}{(9,7)\in R} & \frac{Evn(9+9)}{(9,9)\in R}
\end{array}$$

$$R = \{(0,0),(0,4),(1,1),(1,3),(1,7),(1,9),(3,1),(3,3),(3,7),(3,9),\\ (4,0),(4,4),(7,1),(7,3),(7,7),(7,9),(9,1),(9,3),(9,7),(9,9)\}$$

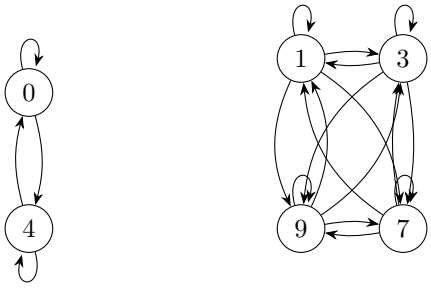
Splits J by parity: evens $\{0,4\}$, odds $\{1,3,7,9\}$.

$j \in J \Rightarrow |j-j| = 0,\ 2 \mid 0 \Rightarrow (j,j) \in R$ — $Rflx(R)$

$(j,k) \in R \Rightarrow 2 \mid |j-k| = |k-j| \Rightarrow (k,j) \in R$ — $Sym(R)$

$(j,k),(k,l) \in R \Rightarrow j,k$ same parity, k,l same parity $\Rightarrow j,l$ same parity $\Rightarrow (j,l) \in R$ — $Trns(R)$

$$Rflx(R) \wedge Sym(R) \wedge Trns(R) \ \Rightarrow\ Eqv(R)$$



Two complete cliques (with self-loops) — one on $\{0,4\}$, one on $\{1,3,7,9\}$ — and *no* edges between them. Classes: $\{0,4\}$, $\{1,3,7,9\}$.

(ii) **N** $PO(S)$

2b.ii

$S := \{(e,f) \in E \times E \mid (e \geq f) \vee (e \mid f)\}$, $E = \{0,1,4,9\}$. Exhaustive check of all 16 pairs of $E \times E$:

$$\begin{array}{cccc}
\frac{(0\geq 0)\vee(0\mid 0)}{(0,0)\in S} & \frac{(0\geq 1)\vee(0\mid 1)}{(0,1)\notin S} & \frac{(0\geq 4)\vee(0\mid 4)}{(0,4)\notin S} & \frac{(0\geq 9)\vee(0\mid 9)}{(0,9)\notin S} \\
\frac{(1\geq 0)\vee(1\mid 0)}{(1,0)\in S} & \frac{(1\geq 1)\vee(1\mid 1)}{(1,1)\in S} & \frac{(1\geq 4)\vee(1\mid 4)}{(1,4)\in S} & \frac{(1\geq 9)\vee(1\mid 9)}{(1,9)\in S} \\
\frac{(4\geq 0)\vee(4\mid 0)}{(4,0)\in S} & \frac{(4\geq 1)\vee(4\mid 1)}{(4,1)\in S} & \frac{(4\geq 4)\vee(4\mid 4)}{(4,4)\in S} & \frac{(4\geq 9)\vee(4\mid 9)}{(4,9)\notin S} \\
\frac{(9\geq 0)\vee(9\mid 0)}{(9,0)\in S} & \frac{(9\geq 1)\vee(9\mid 1)}{(9,1)\in S} & \frac{(9\geq 4)\vee(9\mid 4)}{(9,4)\in S} & \frac{(9\geq 9)\vee(9\mid 9)}{(9,9)\in S}
\end{array}$$

$$S = \{(0,0),(1,0),(1,1),(1,4),(1,9),(4,0),(4,1),(4,4),(9,0),(9,1),(9,4),(9,9)\}$$

$$\dfrac{\dfrac{1,4 \in E, \quad 1 \mid 4 \text{ (long division: } 4 = 1 \times 4 + 0), \quad 4 \geq 1}{(1,4) \in S \qquad (4,1) \in S \qquad 1 \neq 4}}{\dfrac{\neg AntSym(S)}{\neg PO(S)}}$$

2c — $R(\varphi) := \{(u,v) \mid \varphi(u) = \varphi(v)\}$**, given** $PO(T, \leq)$

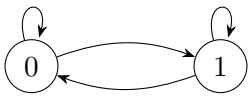
2c

Claim: $\forall \varphi \in Fnc(S,T)\ PO(R(\varphi))$. N.

Rflx and Sym always hold, for any φ (needed before the counterexample below):

$$\dfrac{\dfrac{u \in S, \quad \varphi(u) = \varphi(u)}{(u,u) \in R(\varphi)}}{Rflx(R(\varphi))} \qquad \dfrac{\dfrac{(u,v) \in R(\varphi), \quad \varphi(u) = \varphi(v)}{\varphi(v) = \varphi(u)}}{\dfrac{(v,u) \in R(\varphi)}{Sym(R(\varphi))}}$$

Counterexample: take φ *constant*, $\varphi(x) := 0$ for all $x \in S$ (valid since $T \neq \{\}$; label the constant value 0). Since $\nu(S) \geq 2$, let $0, 1$ denote two distinct elements of S (relabeling if necessary):



$$\dfrac{\dfrac{\dfrac{\dfrac{\varphi(0) = 0 = \varphi(1)}{(0,1) \in R(\varphi) \qquad (1,0) \in R(\varphi) \qquad 0 \neq 1}}{\neg AntSym(R(\varphi))}}{\neg PO(R(\varphi))}}{\neg\big(\forall \varphi \in Fnc(S,T)\ PO(R(\varphi))\big)}$$

3a — P and Q (divisibility orders)

$J = \{0, 1, 3, 4, 7, 9\}, \quad E = \{0, 1, 4, 9\}.$

(i) $\boxed{\mathbf{Y}}$ $PO(P)$**,** $P := \{(j, k) \in J \times J \mid j \mid k\}$

3a.i

Exhaustive check of all 36 pairs of $J \times J$:

$$\begin{array}{cccccc}
\frac{0 \mid 0}{(0,0) \in P} & \frac{0 \nmid 1}{(0,1) \notin P} & \frac{0 \nmid 3}{(0,3) \notin P} & \frac{0 \nmid 4}{(0,4) \notin P} & \frac{0 \nmid 7}{(0,7) \notin P} & \frac{0 \nmid 9}{(0,9) \notin P} \\
\frac{1 \mid 0}{(1,0) \in P} & \frac{1 \mid 1}{(1,1) \in P} & \frac{1 \mid 3}{(1,3) \in P} & \frac{1 \mid 4}{(1,4) \in P} & \frac{1 \mid 7}{(1,7) \in P} & \frac{1 \mid 9}{(1,9) \in P} \\
\frac{3 \mid 0}{(3,0) \in P} & \frac{3 \nmid 1}{(3,1) \notin P} & \frac{3 \mid 3}{(3,3) \in P} & \frac{3 \nmid 4}{(3,4) \notin P} & \frac{3 \nmid 7}{(3,7) \notin P} & \frac{3 \mid 9}{(3,9) \in P} \\
\frac{4 \mid 0}{(4,0) \in P} & \frac{4 \nmid 1}{(4,1) \notin P} & \frac{4 \nmid 3}{(4,3) \notin P} & \frac{4 \mid 4}{(4,4) \in P} & \frac{4 \nmid 7}{(4,7) \notin P} & \frac{4 \nmid 9}{(4,9) \notin P} \\
\frac{7 \mid 0}{(7,0) \in P} & \frac{7 \nmid 1}{(7,1) \notin P} & \frac{7 \nmid 3}{(7,3) \notin P} & \frac{7 \nmid 4}{(7,4) \notin P} & \frac{7 \mid 7}{(7,7) \in P} & \frac{7 \nmid 9}{(7,9) \notin P} \\
\frac{9 \mid 0}{(9,0) \in P} & \frac{9 \nmid 1}{(9,1) \notin P} & \frac{9 \nmid 3}{(9,3) \notin P} & \frac{9 \nmid 4}{(9,4) \notin P} & \frac{9 \nmid 7}{(9,7) \notin P} & \frac{9 \mid 9}{(9,9) \in P}
\end{array}$$

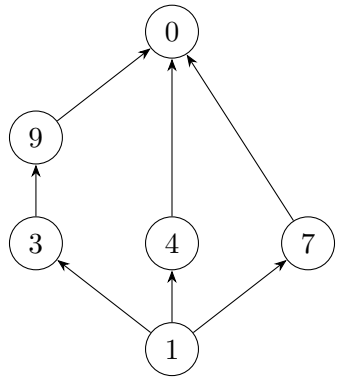
$$P = \{(0,0), (1,0), (1,1), (1,3), (1,4), (1,7), (1,9), (3,0), (3,3), (3,9), (4,0), (4,4), (7,0), (7,7), (9,0), (9,9)\}$$

$$\mathit{Rflx} : j = j \times 1 \Rightarrow j \mid j \qquad \mathit{AntSym} : j \mid k \wedge k \mid j \Rightarrow j = k \qquad \mathit{Trns} : j \mid k \wedge k \mid l \Rightarrow j \mid l$$

Cover relation, verified by division ($a \mid b$ iff $b \div a$ has remainder 0):

$$1 \mid 3\,(3{=}1{\times}3{+}0), \quad 1 \mid 4\,(4{=}1{\times}4{+}0), \quad 1 \mid 7\,(7{=}1{\times}7{+}0), \quad 3 \mid 9\,(9{=}3{\times}3{+}0)$$

$$4 \mid 0, \quad 7 \mid 0, \quad 9 \mid 0 \qquad (\text{every } j \text{ divides } 0 \text{ since } 0 = j \times 0 + 0)$$



Bottom 1, covered by $3, 4, 7$; 9 covers 3; top 0 covers $4, 7, 9$. Arrows point from the covered (smaller) element up to the covering (larger) element, per HsDgrm convention. (Note: $1 \lessdot 9$ is *not* a direct cover — $1 \mid 3 \mid 9$ gives a longer path, so the edge 1–9 is not drawn.)

(ii) $\boxed{\mathbf{Y}}$ $PO(Q)$**,** $Q := \{(e, f) \in E \times E \mid f \mid e\}$

3a.ii

Exhaustive check of all 16 pairs of $E \times E$ (test label shows $f \mid e$, concluding membership of pair (e, f)):

$$\begin{array}{cccc}
\frac{0 \mid 0}{(0,0) \in Q} & \frac{1 \mid 0}{(0,1) \in Q} & \frac{4 \mid 0}{(0,4) \in Q} & \frac{9 \mid 0}{(0,9) \in Q} \\
\frac{0 \nmid 1}{(1,0) \notin Q} & \frac{1 \mid 1}{(1,1) \in Q} & \frac{4 \nmid 1}{(1,4) \notin Q} & \frac{9 \nmid 1}{(1,9) \notin Q} \\
\frac{0 \nmid 4}{(4,0) \notin Q} & \frac{1 \mid 4}{(4,1) \in Q} & \frac{4 \mid 4}{(4,4) \in Q} & \frac{9 \nmid 4}{(4,9) \notin Q} \\
\frac{0 \nmid 9}{(9,0) \notin Q} & \frac{1 \mid 9}{(9,1) \in Q} & \frac{4 \nmid 9}{(9,4) \notin Q} & \frac{9 \mid 9}{(9,9) \in Q}
\end{array}$$

$$Q = \{(0,0),(0,1),(0,4),(0,9),(1,1),(4,1),(4,4),(9,1),(9,9)\}$$

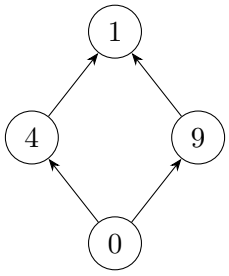
Same three steps, with the divisibility direction flipped ($f \mid e$ instead of $e \mid f$):

$$\mathit{Rflx}(Q) \quad \mathit{AntSym}(Q) \quad \mathit{Trns}(Q) \;\Rightarrow\; \mathit{PO}(Q)$$

Cover relation: an edge (x, y) of Q (meaning $x \leq y$) requires $y \mid x$. Checking all pairs of $E = \{0, 1, 4, 9\}$:

$$4 \mid 0\,(0{=}4{\times}0{+}0) \Rightarrow 0 \leq 4, \qquad 9 \mid 0\,(0{=}9{\times}0{+}0) \Rightarrow 0 \leq 9$$

$$1 \mid 4\,(4{=}1{\times}4{+}0) \Rightarrow 4 \leq 1, \qquad 1 \mid 9\,(9{=}1{\times}9{+}0) \Rightarrow 9 \leq 1$$



Bottom 0, covered by $4, 9$; top 1 covers $4, 9$. Arrows point from covered to covering element. *(Since E has no element like 3 between 1 and 4 or 9 under this order, both 0–4–1 and 0–9–1 are genuine direct covers, unlike part (i). Double-check against the pairwise table above: Q places 0 at the bottom and 1 at the top, since $(0,1) \in Q$ but $(1,0) \notin Q$ — a solved-exam sample that draws this Hasse diagram with 0 at the top has it backwards.)*

3b — symmetric-difference of equivalence relations

Given a set S and $E, F \in \mathit{EqvRlns}(S, S)$.

(i) **N** $\mathit{Eqv}(E \setminus F)$

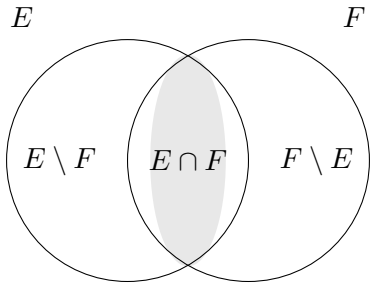
3b.i

$$\frac{\begin{array}{c} F \in \mathit{EqvRlns}(S,S) \\ \hline \mathit{Rflx}(F) \\ \hline \Delta_S \subseteq F \\ \hline \forall x \in S,\ (x,x) \in F \\ \hline \forall x \in S,\ (x,x) \notin E \setminus F \\ \hline \neg\mathit{Rflx}(E \setminus F) \end{array}}{\neg\mathit{Eqv}(E \setminus F)}$$

(ii) **N** $\mathit{Eqv}\big((E \setminus F) \cup (F \setminus E)\big)$

3b.ii

$$\frac{\begin{array}{c} E, F \in \mathit{EqvRlns}(S,S) \\ \hline \Delta_S \subseteq E \qquad \Delta_S \subseteq F \\ \hline \Delta_S \subseteq (E \cap F) \\ \hline \forall x \in S,\ (x,x) \in E \cap F \\ \hline \forall x \in S,\ (x,x) \notin (E \setminus F) \cup (F \setminus E) \\ \hline \neg\mathit{Rflx}\big((E \setminus F) \cup (F \setminus E)\big) \end{array}}{\neg\mathit{Eqv}\big((E \setminus F) \cup (F \setminus E)\big)}$$



every diagonal pair (x, x) lives in $E \cap F$ — the shaded middle

The diagonal Δ_S always sits in the overlap $E \cap F$ (both are reflexive), so it can never appear in $E \setminus F$, $F \setminus E$, or their union — reflexivity fails every time, for every S, E, F.

3c — counting on $B = \{0, 1, 4, 9\}$, $\nu(B) = 4$

3c

$\nu(\mathit{TtlOrdrs}(B)) = 4! = 24$ — **long multiplication**

$$4! = 4 \times 3 \times 2 \times 1$$

$$4 \times 3 = 12, \qquad 12 \times 2 = 24, \qquad 24 \times 1 = 24$$

$\mathit{TtlOrdrs}(B)$: all 24 permutations of $\{0, 1, 4, 9\}$ read as chains:

0<1<4<9	0<1<9<4	0<4<1<9	0<4<9<1
0<9<1<4	0<9<4<1	1<0<4<9	1<0<9<4
1<4<0<9	1<4<9<0	1<9<0<4	1<9<4<0
4<0<1<9	4<0<9<1	4<1<0<9	4<1<9<0
4<9<0<1	4<9<1<0	9<0<1<4	9<0<4<1
9<1<0<4	9<1<4<0	9<4<0<1	9<4<1<0

$\nu(\mathit{EqvRlns}(B)) = \nu(\mathit{Prtns}(B)) = B_4 = 15$ — **via Stirling numbers of the 2nd kind**

$B_4 = \sum_{k=0}^{4} S(4, k)$, where $S(n, k)$ = number of ways to split n labeled elements into exactly k nonempty unlabeled groups, via the recurrence $S(n, k) = kS(n-1, k)+S(n-1, k-1)$, base $S(0, 0) = 1$, $S(n, 0) = S(0, k) = 0$ otherwise.

$n\backslash k$	0	1	2	3	4
0	1	0	0	0	0
1	0	1	0	0	0
2	0	1	1	0	0
3	0	1	3	1	0
4	0	1	7	6	1

Sample recurrence steps: $S(3, 2) = 2 \cdot S(2, 2) + S(2, 1) = 2(1) + 1 = 3$. $S(4, 2) = 2 \cdot S(3, 2) + S(3, 1) = 2(3) + 1 = 7$. $S(4, 3) = 3 \cdot S(3, 3) + S(3, 2) = 3(1) + 3 = 6$.

$$B_4 = S(4, 0) + S(4, 1) + S(4, 2) + S(4, 3) + S(4, 4) = 0 + 1 + 7 + 6 + 1 = 15$$

$\mathit{Prtns}(B)$, all 15 (grouped by shape — $S(4, 1) = 1$ way to make one group of 4, $S(4, 2) = 7$ ways to split into two groups, $S(4, 3) = 6$ ways into three groups, $S(4, 4) = 1$ way into four singletons):

{0,1,4,9} *(1 way, all together)*

{0},{1,4,9} {1},{0,4,9} {4},{0,1,9} {9},{0,1,4} *(4 of the 7 two-group ways: one singleton)*
{0,1},{4,9} {0,4},{1,9} {0,9},{1,4} *(remaining 3 of the 7: two pairs)*

{0},{1},{4,9} {0},{4},{1,9} {0},{9},{1,4} {1},{4},{0,9} *(4 of 6)*
{1},{9},{0,4} {4},{9},{0,1} *(remaining 2 of the 6 three-group ways)*

{0},{1},{4},{9} *(1 way, all separate)*

Count check: $1 + 4 + 3 + 4 + 2 + 1 = 15$. ✓

$\nu(PrtOrdrs(B)) = 219$

This one does *not* reduce to a short hand computation — the number of labeled partial orders on n elements follows the sequence $1, 1, 3, 19, 219, 4231, \ldots$ for $n = 0, 1, 2, 3, 4, 5$ (each term counts many structurally different Hasse-diagram shapes, with no simple closed-form product/sum like $4!$ or the Stirling-number formula above). The value 219 for $n = 4$ is confirmed by exhaustive brute-force check (testing all 2^{16} candidate relations on a 4-element set for reflexivity+antisymmetry+transitivity) rather than hand arithmetic. If your notes give a specific counting formula/recursion for this course, use that method and cite it; otherwise it's acceptable to state the value and reference the sequence.